

Assignment 5

PITWALL - Bilingual Interactive Dashboard

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1. Designer

INFORMATION	DETAILS
Name	Aditya Baidur
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Course	SEG3525 - Design and Analysis of User Interfaces
Prototype	PITWALL - F1 Race Pace, Decoded

2. Dashboard - Objective and Data

DOMAIN AND OBJECTIVE

PITWALL is a Formula 1 race-pace analysis dashboard. It is intended for race fans who want to understand how a driver's pace develops during a race and how that performance compares with the rest of the field. The interface covers four Grands Prix from the 2025 season: Abu Dhabi, Monaco, Great Britain at Silverstone, and Italy at Monza.

DATA SOURCE AND PREPARATION

The historical data comes from the [OpenF1 API](#), an open, unofficial source of Formula 1 data. The dashboard uses session 9839 for Abu Dhabi, 9979 for Monaco, 9947 for Silverstone, and 9912 for Monza. The sporting data is not synthetic and was not generated by AI.

I extracted and integrated a focused subset of the source data: driver name and acronym, team and team colour, fastest lap, fastest-lap number, average lap time, top speed, and selected lap times. I then prepared the data as follows:

- eight drivers are included in each race comparison;
- pit-out laps, missing values, and laps longer than 110 seconds are excluded;
- the pace trace is sampled at approximately five-lap intervals;
- durations are normalized to seconds in `lib/race-data.ts` and formatted for display.

INSPIRATION AND MOCKUPS

The interface does not reproduce an existing dashboard. Its visual hierarchy is informed by motorsport timing conventions: compact data, monospaced numerals, red as a signal colour, and direct driver comparisons. I explored these ideas through two [Figma mockups](#), then simplified the final interface to improve clarity.

3. Design Reflection

A. CHARTS

Line chart - pace development. The line chart shows one driver's lap time over the course of a race. A line chart is appropriate because the data forms a time series and the continuous shape makes trends, stint changes, and faster laps easy to identify. The horizontal axis represents lap number and the vertical axis represents lap time. Lower values indicate faster performance.

Horizontal bar chart - field comparison. Eight drivers are sorted and compared by fastest lap, average lap, or top speed. Horizontal bars support accurate category comparisons while leaving room for driver acronyms. The order and scale update when the selected metric changes.

Together, the charts answer two complementary questions: how did the performance develop, and where does the selected driver rank against the field?

B. THE 3CS

Context. Each chart has a descriptive title, a concise explanation, clearly identified units, controls positioned beside the content they affect, and tooltips for exact values. Summary cards establish the selected driver, fastest lap, top speed, and pace gap. A separate reading note explains the data cleaning and sampling process.

Clutter-free. The dashboard displays only one pace line at a time. Permanent point markers and redundant legends were removed; a point appears only during hover. Grid lines are light and used in one direction. Sampling the pace trace and limiting the comparison to eight drivers prevents unusual events and chartjunk from obscuring the main pattern.

Contrast. Dark charcoal text on white provides strong readability. Racing red is reserved for important headings, references, and active states. The pace line uses the selected driver's team colour. In the bar chart, the selected driver is red while the remaining drivers are grey. Colour is not the only signal: names, values, button states, and tooltips communicate the same information.

C. LAYOUT, TITLE, AND INTERACTIONS

The title, "Race pace, decoded / Le rythme de course, décodé," states the dashboard's purpose directly. A large introduction identifies the race and circuit, followed by summary metrics, the two charts, a written interpretation, and the data source. This sequence moves from general context to detailed analysis.

The user can:

- select one of four Grands Prix, updating the circuit, summary metrics, and both charts;
- select one of four featured drivers for the pace chart;
- switch the comparison chart among fastest lap, average lap, and top speed;
- hover over either chart to inspect exact values;
- switch between English and French at any time.

Selected buttons use clear contrast and an explicit aria-pressed state. Choosing a different race also resets the driver to a valid selection, preventing inconsistent data from remaining on screen.

D. INTERNATIONALIZATION

The selected languages are Canadian English and Canadian French. The English content served as the source. OpenCode helped produce an initial French translation, which I reviewed and adjusted for motorsport terminology. For example, *race pace* is translated as "rythme de course," *best lap* as "meilleur tour," and *race readout* as "lecture de course."

The primary challenge was text expansion in French. Labels such as "vitesse maximale" and "comparaison du peloton" require more space than their English equivalents. Control groups can therefore scroll horizontally on narrow screens, cards allow text wrapping, and the mobile language buttons use EN and FR. Proper names, driver acronyms, and the international unit km/h remain unchanged.

Localization covers the title, navigation, descriptions, summary cards, controls, axes, tooltips, help content, sources, accessible text, and footer. Intl.DateTimeFormat and Intl.NumberFormat apply either en-CA or fr-CA, including localized month names and decimal separators. The document's lang attribute also changes with the selected language to support assistive technology.

4. High-Fidelity Prototype

The prototype is built with Next.js 16, React 19, TypeScript, Recharts, and Tailwind CSS. I selected an editorial motorsport direction rather than a generic analytics template: white surfaces, charcoal typography, restrained racing-red accents, large display type, and monospaced numerals associated with timing data.

Visual hierarchy is created through a large headline, short uppercase labels, grouped cards, and deliberate negative space. Proximity connects each control to the chart it changes. Similar card and button treatments establish consistency. The dark “Race readout” panel creates a distinct figure for verbal interpretation, allowing the written and visual communication to support each other.

The prototype is responsive across mobile and desktop layouts. It also includes a skip link, ARIA labels, screen-reader text versions of chart data, and support for prefers-reduced-motion. These decisions support system-status visibility, consistency, error prevention, recognition rather than recall, and accessibility.

- Portfolio: <https://d1.byaditya.com>
- Hosted prototype: <https://seg-3525-devoir-5.vercel.app>
- Figma mockups: <https://seg-3525-devoir-5.vercel.app/figma>

5. Code

The source code is publicly available on GitHub:

- https://github.com/ABUO023/SEG_3525-Devoir_5

The project separates race data in `lib/race-data.ts`, the main React dashboard in `components/race-dashboard.tsx`, reusable interface components, and global styles. This organization makes the data, translations, and interactions easier to inspect and maintain.

6. Generative AI Acknowledgement

HIGH-FIDELITY PROTOTYPE

I used **OpenCode**, a generative-AI programming assistant, during development. My interactions asked it to analyze the assignment requirements, propose a React and TypeScript structure, integrate the Recharts visualizations, organize English and French interface copy, add additional circuits, and improve accessibility and responsive behaviour. I selected the domain and data, directed the design, revised the results, tested the interactions, and remained responsible for the final implementation. OpenCode did not generate the sporting data; those values come from OpenF1.

USE OF GENERATIVE AI

I used GPT-5.6 and Google Stitch as a supporting tool while preparing this report. It assisted me in reviewing the assignment requirements, organizing evidence from the repository and hosted prototype, and improving the clarity and structure of the report. I independently reviewed and verified all claims, links, session numbers, and connections between the written report and the final prototype before submission.